Do WTO Disputes Actually Increase Trade?

Stephen Chaudoin, University of Illinois at Urbana-Champaign

Jeffrey Kucik, Princeton University

Krzysztof Pelc, McGill University

Abstract

Exporters, trade lawyers, policy makers, and academic literature perceive the WTO's Dispute Settlement Understanding as an important, though costly, venue for facilitating the removal of harmful barriers to trade. If this conventional wisdom is true, then disputes should increase trade. We cast doubt on this possibility through a careful analysis of trade flows in the wake of WTO disputes. We find that WTO disputes do not increase the respondent country's imports of the products at issue. We find only very narrow effects of disputes based on the dispute outcome and issue-area. Although there is significant variation across countries in their responsiveness to disputes, no common explanation seems able to account for this variation. This article highlights and begins to fill a significant gap in our understanding of the purpose of the WTO and its effects on trade.

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The World Trade Organization's (WTO) Dispute Settlement Understanding (DSU) plays an essential role in liberalizing international trade. This belief has risen to the status of assumption, as the literature has anointed the DSU as the "backbone",[1] the "cornerstone",[2] and the "centerpiece"[3] of the multilateral trade regime. Outside of the trade literature, the DSU has come to represent a uniquely effective international court whose rulings are followed by sovereign nations.[4]

We ask a simple question: Do WTO disputes increase trade? More specifically, do respondents import more of the products at issue after a dispute than before? The belief that litigation leads to liberalization implies an affirmative answer. The DSU is designed as a mechanism for countries to compel their trade partners to remove harmful barriers to trade. It is widely seen an important tool that a complainant country can use to open unfairly protected markets in the respondent country. The Most Favored Nation (MFN) principle, which extends into every part of dispute settlement, ensures that every concession offered to one member is also offered to all others.[5] Policy adjustments made by a respondent country in response to an unfavorable ruling should increase market access for the complainant as well as all other countries harmed by the offending barrier. This role of the DSU has been cited as one of the main reasons why countries join the WTO in the first place.[6] This role is also one of the main reasons why exporting firms pressure their governments to initiate trade disputes.[7] This role is apparent in more theoretical arguments about the role of dispute settlement mechanisms more broadly.[8] Some observers have even suggested that the success of the DSU can fill in where multilateral negotiations have failed.[9]

We show that the evidence falls short of offering unambiguous support for the belief that WTO

[1] The former Director General of the WTO, Mike Moore, declared the DSU to be the "backbone of the multilateral trading system" Busch and Reinhardt (2000).
[2] See Trachtman 1999.
[3] See Lichtenbaum 1997.
[4] See, for instance, Goldstein et al. 2000; Alter 2003.
[5] Bagwell and Staiger (2004).
[6] Cattaneo and Primo Braga (2009).
[7] Davis (2011).
[8] Mansfield, Milner and Rosendorff (2002).
[9] Steinberg (2009).

disputes increase the respondent's imports. We analyze detailed data on imports values at the country-product-year level from 1995-2010. Since disputes are most often about specific products, as opposed to broad sectors, we use the most disaggregated product-level data available for all countries in the sample—6-digit Harmonized System (HS) level. We then code data describing the occurrence of a dispute against that particular country (the respondent) over that particular product. We also code relevant characteristics of the dispute, such as its outcome, issue area, as well as characteristics of the respondent.

In models that account for unobserved country-year and country-product heterogeneity, the value of the respondent's imports of the disputed products is only modestly higher after a dispute, compared to the value before the dispute. Specifically, a dispute increases import values by less than 10%. This effect is not statistically significant. These null results are also apparent in non-parametric analysis of import values before and after disputes, in models that attempt to detect lagged or delayed effects of disputes, and in a variety of other robustness checks.

This article is first and foremost a careful empirical analysis. Our goal is to assess an empirical pattern that is implied by conventionally held beliefs about the DSU. After presenting these initial findings, we push forward by examining the heterogeneity of dispute effects. We conduct an extensive analysis of whether dispute characteristics, such as the outcome or issue-area of the dispute, and respondent characteristics, such as the identity or features of the country targeted by the dispute, enhance or mute the effect of disputes on trade. Disputes that are withdrawn are associated with greater increases in trade, while those ending with a ruling or a mutually agreed solution do not have a positive effect on trade. Looking at dispute issue-areas, disputes over safeguards appear to result in greater trade *ex post*, while disputes over SPS, antidumping, and agriculture do not.

We also find considerable variation across respondents in the degree to which disputes increase

trade. While disputes against certain countries, such as Venezuela, India and Argentina, have resulted in increased trade, disputes against many other states, such as Korea and Mexico, have had negligible or even negative effects on trade. When looking at respondent characteristics as explanations for this variation, we find that disputes against countries experiencing economic and financial crises are less successful at increasing trade, while few other country-level characteristics give strong explanations for why some respondents are more responsive to trade disputes than others. These results concerning heterogenous dispute effects point future researchers towards where they should (or perhaps should not) look to explain the conditions under which disputes increase trade.

Our findings have several implications for the overall assessment of the role of dispute settlement in global trade governance. First, our findings could be evidence that dispute settlement does not succeed in returning trade to where it would have been absent the respondent's WTO-violating policy. The initial barrier to trade may alter production patterns or other market decisions to such a degree that trade does not return to pre-violation patterns even once the offending barrier has been disputed. Disputes may also simply be failing to induce the removal of harmful trade barriers. Second, our results, combined with other recent results, could be evidence that disputes cause trade diversion, not trade creation. Using dyadic, sectoral level data, Bechtel and Sattler (2013) find that respondents import approximately $US 7.7 billion more from the complainant during the first three years after the dispute. A positive dyadic finding combined with a negative or null monadic finding is evidence of trade diversion. If we believe that disputes are increasing trade for the litigants but not for the WTO membership as a whole, then this is evidence that the trade gains for the litigants are being offset by diversion away from non-litigants. Lastly, the DSU could act as a deterrent against future trade barriers or a forgiveness mechanism for certain defections.[10] Theoretical arguments about escape or deterrence imply that the benefit of

[10] E.g., Rosendorff (2005).

a dispute to the complainant should be found either in direct compensation to the complainant by the respondent or by the respondent forgoing future tariff barriers. Since our analysis shows that one of the most likely places to look for the direct effects of dispute settlement- an increase in imports into the respondent- falls short of expectations, these other channels represent important lines for future research to quantify the effects of disputes.

1 Why Disputes Ought to Increase Trade

Why should we expect WTO disputes to increase trade? The DSU is designed as a forum where one country can try and legally compel the removal of another country's barrier to trade. A complainant country (or countries) requests consultations over an allegedly WTO-noncompliant trade policy of a respondent country. The dispute participants try to negotiate a solution, and if they cannot agree, a DSU panel is established to adjudicate the dispute and issue a ruling. If the complainant wins this challenge, the respondent either compensates the complainant for damages or removes the offending trade barrier.[11] Often, the objectionable policies in question are protectionist barriers designed to protect import-competing firms in the respondent country.

Any concession should affect all members, not just the dispute participants. Indeed, Article 3.5 of the DSU, which reads that "all solutions (...) shall be consistent with [the WTO] agreements," implies that even bilateral deals reached in private consultations must be MFN-compliant, and thus extend any trade concessions to all WTO members.[12] In a legal sense, every DSU dispute is a "class action" in which each member harmed by the respondent's illegal barriers to trade should benefit from the outcome of litigation.[13] Further, the growing importance of third party status, through

[11] The original purpose and design of dispute settlement is to settle disputes, rather than act as a balancing mechanism, where violations are offset through compensation (Jackson, 2004; Hudec, 1993).

[12] For an examination of how WTO rules seek to prevent "bilateral opportunism" in multilateral negotiations and dispute settlement, see Bagwell and Staiger (2004).

[13] It is for this reason that scholars argue poorer countries are able to free-ride on liberalizing dispute settlement activity by more developed countries (Bown, 2009, 2005).

which non-litigant member countries may join a dispute's proceedings, ensures that interested parties can observe and contribute to proceedings at every stage. Therefore, dispute settlement in trade is not a purely bilateral exercise, as it is in other issue-areas.[14]

Existing literature highlights several reasons why a WTO dispute, or dispute settlement mechanisms in general, can facilitate compliance, i.e., the removal of illegal trade barriers. Disputes can provide political cover for the respondent to remove the trade barrier. The dispute can also raise the costs to respondent of leaving the barrier in place.[15] There are many sources, domestic and international, of these noncompliance costs. The WTO is fundamentally a diplomatic organization, and countries concede to legal challenges today partly out of the expectation that they may wish to challenge others' policies in the future. Such a systemic effect was already present in the (albeit legally weaker) GATT system, in which countries frequently conceded even though they had the option to block proceedings against them.[16] The normative condemnation flowing from a definitive finding of violation is itself expected to compel compliance.

Furthermore, the final remedy in every dispute is retaliation: a complainant that has not obtained compliance from the respondent can legally suspend concessions, and raise tariff barriers to the level of injury sustained by the violation. This effect is potent enough that vulnerability to such retaliation, measured by the amount of trade exposed to suspension of concessions by the complainant, has been shown to explain whether a respondent concedes or not during litigation.[17] This bilateral pressure is often multilateral in nature, since the WTO facilitates the participation of other countries in disputes: members can join as co-complainants, or as third parties, in which case they are in the room during both consultations and litigation.

Disputes can also activate domestic actors who punish their government politically for offering

[14] Compare this with international investment and business arbitration, where dispute settlement proceedings are entirely private, and usually strictly bilateral. In those cases, there is no talk of free-riding by other actors.

[15] Busch and Reinhardt (2000); Bello (1996); Bagwell and Staiger (2005); Bown (2009); Maggi (1999); Maggi and Staiger (2011).

[16] Busch and Reinhardt (2005, 2000).

[17] Bown (2009).

distortionary protection to industries. Mansfield, Milner and Rosendorff (2002) argue that this *ex-post* cost function is the very reason for which countries join trade agreements to begin with: as a means of credibly conveying to voters and consumers that the government is not engaging in distortionary protection. The implication is that voters are able to sanction leaders who go against these promises.

Existing literature further supports the idea that WTO disputes can compel the removal of harmful trade barriers with other pieces of evidence. For example, Davis (2011) argues that domestic firms pressure their government to initiate WTO disputes in order to facilitate greater market access. Legal firms advertise their ability to increase market access by taking down illegal trade barriers.[18] And governments similarly boast that their litigation efforts open up markets and increase trade.[19]

If disputes truly are a means of getting the respondent to remove harmful trade barriers, then the empirical prediction is straightforward: *in the wake of disputes, the respondent should import the products involved in the dispute in greater amounts.* This is *not* to say that every dispute succeeds or that every dispute is associated with significant increases in imports. Rather, existing arguments predict that, on average, disputes should lead to increased imports.

To our knowledge, the only two existing articles addressing a similar question to ours are Bown (2004) and Bechtel and Sattler (2013). Bown analyzes the factors that cause disputes to lead to greater or lesser increases in imports from the complainant, covering the years 1973-1998. Notably, and consistent with the findings we present below, the average change in log imports after a dispute in his sample is close to zero, only 0.0078. Bown also finds that the threat of complainant retaliation

[18] From an illustrative statement by the trade law firm Cassidy Levy Kent: "[We] advise companies, groups of companies, and governments on means to secure market access through the dispute settlement process under the WTO." From company homepage, Last accessed on July 4th, 2014: `http://www.cassidylevy.com/practice-areas/wto`

[19] As a representative statement, see the USTR press release about its challenge of China's Treatment of U.S. Suppliers of Electronic Payment Services: "Removal of the monopoly that China has provided to China UnionPay would create significantly expanded business opportunities in China's huge and growing market." USTR Requests WTO Dispute Settlement Panels in Two Cases Against China, 2011, Last accessed on July 4th, 2014: `http://www.ustr.gov/about-us/press-office/press-releases/2011/february/ustr-requests-wto-dispute-settlement-panels`

is an important factor driving the degree of respondent liberalization.

Bechtel and Sattler (2013) use dyadic data at the sectoral level (SITC 1-digit level) and find that, in the three years following a dispute, respondents import approximately $7.7 billion more from complainants and third parties who side with the complainant. This estimate is very high, especially given that the majority of trade disputes cover trade flows valued at much less than $7.7 billion. For frame of reference, Bown and Reynolds (2014) analyze trade flows before and after WTO disputes and the types of policies that precipitated the dispute. They find that for disputes which challenge changes to the respondent's trade policies, the median value of trade in the year preceding the WTO-illegal policy change is only $57 million. For the WTO dispute to yield increases in trade of $7.7 billion, the dispute would have to result in an increase in imports by a factor of over 100.

Our analysis builds off of these findings in two ways. We focus on specific products named in the dispute, as opposed to the sectors in which they are found. This is important because WTO disputes are often about policies governing specific products, like an antidumping duty levied against specific imported goods, rather than policies that affect broad sectors of goods. Our data thus more closely match the scope of each dispute. Second, we pay particular attention to variation in dispute features that are thought to be important, such as the outcome of the dispute, the issue-areas covered in the dispute, and characteristics of the respondent. In the sections that follow, we motivate in greater detail why these dispute characteristics may have an effect on the degree to which a dispute increases imports.

1.1 Legal Outcome

What types of factors might make disputes more or less effective at increasing imports? The effects of disputes on trade may vary as a function of their outcome. There are three ways in which a

dispute can be concluded: the complainant may drop the case; the parties can reach a "mutually agreed solution" (MAS); or the dispute may result in a panel ruling, which may or may not be appealed. In between a dispute's initiation and its ultimate outcome, we can also think of a dispute as being "ongoing."

Dropped Cases. No less than 35% of all WTO disputes end up as "dropped" cases, defined as disputes in which no formal settlement is notified to the DSU, and no panel is established. These disputes often are left out of the best known analyses of WTO dispute settlement.[20] Yet recent estimates suggest that as much as *half* of dropped disputes actually reach some settlement,[21] which is never notified to the membership, as WTO rules require parties to do. This was the case in the first WTO dispute ever filed, where Singapore subsequently dropped its case against Malaysia before a panel being formed. Only later did it become clear that a settlement had actually been reached, as Malaysia stated in a DSU meeting that it "appreciated the cooperative spirit and understanding of Singapore's delegation in coming to an amicable solution on this matter."[22] Absent such an announcement, the case would be perceived today as having been abandoned by Singapore.

Often, dropped cases linger for years in the consultation stage without leading to the formation of a panel. Examples include *US—Import Prohibition of Certain Shrimp and Shrimp Products* (DS61), brought by the Philippines against the US, or *Argentina—Wheat Gluten* (DS145), filed in response to countervailing duties the EU placed on Argentine wheat. It is often impossible to ascertain why these cases are abandoned, precisely because no formal notification is provided to the WTO. It may be that the WTO-illegal policy is dismantled by the respondent in response to the case's filing, rendering formal dispute settlement unnecessary. If merely bringing a case forward is

[20] Busch and Reinhardt (2005).
[21] Reynolds (2009)
[22] WTO Dispute Settlement Body, Minutes of Meeting Held in the Centre William Rappard on 24 April 1996. WTO Doc WT/DSB/M/15.

enough to affect policy change in respondent states, then dropped cases may signal concessions by the offending party. It may also be that settlements go unnotified to allow the parties to reach more cooperative arrangements, away from domestic pressures. As we argue below, pre-trial settlements allow complainants to tailor mutually agreed solutions to their specific needs; disputes that appear as dropped cases may render this even more likely. Under these conditions, disputes that are initiated, but that reach neither a formal settlement nor a panel ruling, might lead to an increase in post-filing trade flows.

Conversely, there may be other reasons that cases get dropped, which would imply different effects of the dispute on trade. Complainants may simply lack the resources required to take the dispute forward. A complainant may file a dispute to signal dissatisfaction but lack either the legal resources or the political will required to push the case onward. Complainants may also receive new information during consultations about the relative merits of their case or about the respondent's resolve that could make litigation unfavorable. Any of these explanations suggest that cases get dropped because the complainant walks away for reasons other than the respondent conceding. In those situations, we would not expect to observe an increase in trade flows. However, theory suggests that complainants abandoning a dispute outright should not be the predominant explanation for dropped cases. The significant costs associated with litigation, together with a likely reduction in perceived resolve in subsequent filings, cast doubt on the belief that complainants would file disputes from which they are likely to walk away empty-handed.

Mutually Agreed Solutions. Mutually agreed solutions are slightly less common, occurring in about 20% of all merchandise disputes, e.g. *Japan—Procurement of a Navigation Satellite* (DS73), brought by the EC, and *Slovakia—Safeguard Measure on Imports of Sugar* (DS235), brought by Poland.[23] MAS refers to cases in which the parties to the dispute have settled the issue

[23]We count disputes as ending in an MAS if they were settled prior to the formation of a panel. Some official notifications of MAS to the DSU refer to cases about which there were protracted compliance proceedings in the wake of a failed appeals process. Cases of that types, such as DS103 are coded as cases where a ruling occurred.

during consultations rather than a panel decision. MAS ought to have a less theoretically ambiguous effect on trade. It should lead to an increase in trade because MAS, by definition, implies (at least some) compromise on the part of the respondent. There is little reason to expect that complainants announce a settlement without having secured some gain. The literature on courts also leads us to expect that by saving the costs of litigation they would incur if the case were to go to panel, parties should be willing to concede relatively more.[24] As a result, under a MAS, the respondent ought to grant more access to their domestic market, and extending any concessions to other countries through MFN, thus leading to increased imports.[25]

Rulings. Approximately half of all disputes proceed to litigation and result in a panel report. Historically, rulings have significantly favored complainants, who win at least some component of virtually every dispute filed. A little over 90% of all merchandise cases are decided in favor of the complainant.[26] This pro-complainant bias is hardly surprising given the high cost of disputes, and the selection process it entails. Assuming that the DSU affects state behavior, respondents ought to dismantle trade barriers that the panel rules against. As a result, on average, rulings ought to lead to greater market access and increased trade.

The prediction that rulings increase trade takes implementation for granted. In practice, we know that states vary in the extent to which they comply with DSU rulings. While not common, compliance proceedings have been initiated in several situations in which there was doubt cast over respondents' implementation of the panel's recommendations. Poor compliance may result from non-random selection into the panel stage. Respondents who see cases through to rulings may be

[24] Gilligan, Johns and Rosendorff (2010).

[25] In some cases, a settlement may result in compensation in some form other than lowered trade barriers. The *US—Cotton* case, filed by Brazil, may be the most salient such example. It also confirms our expectations: in this case, as in others, compensation is seen as an interim measure. The dispute remains on the agenda of the DSU. Compensation is formally seen as a temporary measure, awaiting the conclusion of the US Farm Bill. The expectation on the part of Brazil is that eventually, the dispute will yield abatement of US cotton subsidies, and thus more access into the US market.

[26] We measure a victory as the panel siding with the complainant on at least one of the legal claims made in the case. There is variation over the precise content of rulings. Complainants often do not win every claim they make, and many claims are not ruled on directly by the panel/appellate body at all. However, complainants win at least some portion of the dispute almost invariably.

particularly unwilling to compromise on their policies. Specifically, rulings may occur most often in those situations in which the respondent is *least likely* to back down.[27] If that is true, then rulings may not have a significant impact on trade flows between members.

Two remarks are in order. First, while it is difficult to measure compliance, the existing data show that implementation rates are higher than we might expect.[28] The historical record simply does not show that respondents are non-compliant on average, openly flouting WTO decisions.[29] Second, it may well be the case that the DSU "lacks teeth." Kono (2006) argues that the system is effective, but that it operates primarily through reputation in the international system, rather than through centralized enforcement. In either case, whether compliance hinges on self-enforcement through a concern over reputation, or the fear of trade retaliation, there is good reason to believe that rulings in the highly legalized trade regime should have an effect on country behavior, and thus on trade flows.

1.2 Issue-Area

Disputes also vary in terms of the type of protectionist barrier being challenged. Variation in the content of the dispute is important because some issues areas are particularly contentious and may be less likely to result in substantive policy changes by respondents. For example, agriculture is typically seen as the most politically fraught issue in the WTO. The unwillingness of (especially developed) members to commit to concessions in agriculture is widely acknowledged as a key sticking point in the stagnant Doha Round. As a result, agrarian products are often cited in WTO disputes, but this may be precisely because governments erect entry barriers they refuse to dismantle.

[27] Reinhardt (2000).
[28] Wilson (2007).
[29] It is true that respondents may not grant full concessions. While the evidence suggests compliance, on average, wholesale repeal of entry barriers is not always the result. We take this concern seriously, and the third part of the analysis is designed to explain precisely which kinds of states are more or less responsive to sanction by the DSU.

Another important area of trade law is the WTO's allowance for escape clauses, which allow states to temporarily shirk their obligations. Such measures include antidumping, countervailing duties, and safeguards. In practice, antidumping is used nearly 10 times more often than safeguards. However, both policy tools frequently are targeted in disputes. Of the first 425 WTO disputes, 85 relate to antidumping and 38 relate to safeguards.

To understand why these escape clauses might condition the relationship between disputes and trade, consider the reasons these policies are enacted. Both actions require that a domestic industry petitions its government for protection. In the case of antidumping, firms petition when they perceive that foreign imports are priced unfairly. Governments taking advantage of the WTO's flexibility are, therefore, facing significant domestic political pressure to insulate the market from competition. As a result, even if a government loses a WTO dispute, there might be high domestic political costs from dismantling the barrier.[30] Sanitary and phyto-sanitary (SPS) measures also represent an increasingly salient portion of international trade law, both at the WTO and in the context of regional trade agreement negotiations. Disputes over these issues are potentially thorny, since they touches on sensitive issues, such as genetically modified crops and food safety.

In all of these politically sensitive issue-areas, we might expect that the trade effects of dispute settlement are attenuated by the high domestic costs of dismantling entry barriers. Agricultural subsidies and enforcements of the WTO's flexibility system each result from political pressure that may strengthen the resolve of respondents and, importantly, lead to less post-dispute trade *even if* the respondent loses in court.

[30]Note that this logic points to an alternative way to think about domestic audience costs in the context of the WTO. We traditionally think that governments pay a cost for violating international law, and this ought to create political incentives for office-seeking governments to comply with the DSU. However, for at least some portion of the domestic constituency, violation of the law may be a credible signal of precisely the government behavior they want.

1.3 Respondent's Responsiveness

Disputes also vary, of course, in which country is being targeted as the respondent. We are interested in the possibility that some respondents are more "responsive" than others. In other words, a dispute against one country might yield greater increases in imports into that respondent country compared to a dispute against another country.

Which country characteristics explain responsiveness? Here, we draw on theoretical arguments from existing literature to compile a list of potential respondent characteristics that might moderate, e.g. magnify or mute, the effect of a dispute on imports. First, characteristics of the respondent's domestic market, like the country's wealth might affect how responsive it is to trade disputes. Countries with greater legal capacity have been shown to file more WTO disputes, with a country's per capita income being the most commonly used proxy for legal capacity at the WTO.[31] It is plausible that countries with greater legal capacity may be *less* responsive to WTO disputes. They have greater resources to rely on when defending themselves against complaints. They may also be better able to drag out disputes, and the lengthiness of disputes has been shown to discourage some potential litigants.[32]

Related to the domestic market, countries facing economic crises might also be less responsive to WTO disputes. Crises magnify pressures for import relief from powerful domestic import-competing groups, and countries facing hard times may be less likely to respond to WTO disputes by allowing increased imports. Increased imports threaten employment, a particularly salient concern during hard times.

A country's position in the global market is also likely to make a difference in a dispute's effectiveness.[33] Since the WTO has no central enforcement mechanism, WTO disputes rely on

[31] Busch and Reinhardt (2003); Busch, Reinhardt and Shaffer (2009); Guzman and Simmons (2005); Horn, Mavroidis and Nordström (1999).
[32] Davis and Shirato (2007).
[33] Bown (2004).

the threat of retaliation as a last-resort enforcement device. If the complainant wins a dispute, it can be granted the right to erect retaliatory trade barriers against a noncompliant respondent. A country's trade position thus makes that country more or less responsive to a WTO dispute, since it could me more or less sensitive to potential retaliation. A country reliant on exports has more at stake should other countries obtain authorization to retaliate against it. On the other hand, a country that imports a lot from other countries might be less sensitive to the threat of retaliation, since it imports products that might be harmed by counter-retaliation. Countries with higher imports might be less responsive to WTO disputes.

Regime type might be another factor accounting for members' responsiveness to WTO disputes.[34] If WTO disputes are a mechanism to mobilize domestic actors to oppose protectionism, then this effect should be greater in more democratic countries and lesser in autocratic countries. In autocratic countries, mobilizing pro-free trade groups would have less effect, since they cannot push for policy changes.

Finally, a country's past experience with the DSU might also make them less responsive. As countries gain experience with the complex legal system, they may be better able to defend themselves against the charges levied by the complainant.

2 Data and Models

To analyze whether disputes increase the value of imports into a country, we construct a dataset with one row per country i, for each product s, in each year t, such that the unit of observation is the country-product-year. We start with the 320 merchandise disputes that occurred between 1995 and 2010.[35] There were 31 countries who were respondents in at least one of these disputes. We

[34] Mansfield, Milner and Rosendorff (2002, 2000).

[35] There were 419 disputes between 1995 and 2010 but only 320 are merchandise disputes, i.e. they deal with trade in particular goods. The remainder of disputes related to domestic trade regulation and practices, such as the United States' use of 'zeroing' in the calculation of dumping margins.

then collected data on the products named in these disputes. We gather the data at the 6-digit level of the Harmonized System codes, the highest level of disaggregation that allows for concordance across time. Different disputes name more or fewer products, and some disputes name products at a level of aggregation that is greater than HS-6. For disputes that name goods at higher levels of aggregation—e.g. DS141 citing HS 52 ("Cotton, including yarns and woven fabrics")—we use all 6-digit lines under that larger heading.[36] Our data cover 4,934 distinct products. These include all products named in merchandise disputes from DS1-DS420. The resulting data represent what we believe is the most comprehensive and detailed dataset of dispute-specific product-level trade flows into respondent markets.

2.1 Dependent Variable

Our dependent variable measures the value of imports into the respondent country of a particular product in that year. Specifically, let IM_{ist} be the natural log of the value, in 1996 U.S. dollars, of country i's imports of product s in year t from all exporters.[37]

To examine the effectiveness of disputes, we focus on import *values* as opposed to a particular measure of *policies* for two reasons. First, data on disputed policies like tariffs, standards, and domestic regulations are incomplete for the full set of products, countries, and years covered here. Second, governments' announcement of policy changes are often misleading, since governments have an interest to claim to have reformed their policies even when they have not, or worse still, to substitute the measure at issue with a new measure amounting to a similar barrier. Contrarily to government statements, trade flow data do not lie. Litigants and policy makers should only care about the *de jure* effect of WTO disputes on trade policies insofar as those changes have *de facto*

[36] This coding decision does not affect results. We get similar results limiting the sample only to disputes which named products at the 6-digit level.

[37] Note, there are not missing observations or zero-valued amounts of trade. The analysis, after all, concerns products which were involved in a WTO dispute, so there is at least some positive amount of trade in that product, for every product in the analysis. Thus, usual concerns about how to treat zero values in gravity model approaches do not apply.

effects on trade flows.

2.2 Explanatory Variables

Our analysis employs a series of independent variables. We start with a dichotomous dispute indicator. $Dispute_{ist}$ is coded "1" for all country-product-year observations after a dispute has been initiated. This dichotomous indicator allows us to test whether dispute settlement has a positive effect on imports, on average.

2.2.1 Dispute Status

Since the effect of a dispute on imports is potentially conditioned by the legal outcome of that dispute, we construct a disaggregated measure of dispute "status," which represents the stage that a dispute has reached in a particular importer-product-year.

1. No Dispute: No WTO members have yet requested consultations with the importer regarding that product, as of year t
2. Dispute Dropped: A WTO member has requested consultations with the importer regarding that product, but the dispute has been dropped
3. Mutually Agreed Solution (MAS): Consultations have been requested and both sides have reached a mutually agreed upon solution
4. Panel Ruling: Consultations were requested, a panel was established, and the panel has issued a ruling
5. Dispute Ongoing: Consultations were requested, but the dispute has not been dropped, resulted in a MAS, nor resulted in a panel

These options span the possible statuses for an importer-product-year, and do not overlap with one another.[38] Let $status^{j}_{ist}$ indicate the status of the observation, where $j \in (1, 2, 3, 4, 5)$ correspond to each of the five possible statuses. 2,004 (11.1%) of the country-products concluded with a dropped dispute. 5,584 (31.0%) reached a mutually agreed solution, and 10,437 (57.9%) resulted in a panel ruling.

[38] To be sure, there are interesting subsets to each status, e.g. whether a panel *and an appeals* panel have been established. Here we focus on the most prominent outcomes, reflecting the various stages through which disputes may generally progress.

Given the econometric approach described below, it is relevant to describe the sub-sample of observations pertaining to country-products that do experience a dispute, i.e., excluding all country-products that never experience a dispute. There are approximately 121,000 observations pertaining to country-products that do experience a dispute. Of these, approximately 35% of the observations occur before a dispute. 15% occur after a dispute has been dropped. 22% occur after a mutually agreed solution, and 17% occur after a panel ruling. 11% occur while a dispute is ongoing.

In terms of the full country-product-year sample, the vast majority of observations, 2,121,784 (95%), occur before a dispute or for country-products that do not ever experience a WTO dispute. 21,471 (1%) occur after a dispute has been dropped. 33,526 (1.5%) occur after a mutually agreed solution, and 31,377 (1.4%) occur after a panel ruling. 18,128 (0.8%) observations were coded as belonging to an ongoing dispute.

2.2.2 Dispute Issue-Area

The second set of controls describe features of the dispute. Any effect that a dispute has on trade is likely to be qualified not just by its outcome, but also by the legal issues it raises. We code dichotomous measures of whether the dispute was about one of the two most commonly contested escape clauses in the WTO—antidumping [AD] and safeguards [*Safeguards*].[39] We also code whether the disputed product is an agricultural good [*Agriculture*]. We also code an indicator of whether the dispute involves sanitary and phyto-sanitary (SPS) measures [*SPS*].

2.3 Country-Year Control Variables

We also consider the effects of time-varying characteristics of the respondent. We include four measures relating to each respondent's domestic political institutions, economy, and previous ex-

[39] In practice, antidumping is used ten times more frequently than safeguards, but leads to just over twice as many contestations. In other words, safeguards are contested relatively more frequently.

perience with dispute settlement. The first is a measure of regime type, which could affect a country's propensity for cooperation, and compliance with international legal rulings. $Polity_{it}$ measures country i's combined Polity IV scores in year t. We also include *per capita* income on the right-hand side, since richer countries are more likely to conduct more trade and to have higher levels of legal capacity. $lnpcgdp_{it}$ measures the natural log of country i's per capita income in year t. To account for domestic political pressure, we include a measure of whether a country is in the midst of an economic crisis in the year under observation, using data from Reinhart and Rogoff (2009).

In addition, we include the number of times a country has been a respondent in a WTO dispute. Previous experience with dispute settlement may shape a country's willingness to concede market access as a result of the dispute, or its ability to fend off new challenges. $Respondent_{it}$ measures the (logged) number of times that country i has been a respondent in a WTO dispute, up to year $t-1$. $Crisis_{it}$ codes the number of crises a country is experiencing across six possible dimensions: banking, currency, domestic default, external default, inflation, and stock market crises.[40]

We refer to these control variables collectively as X_{it}. To be sure, these variables are not the only country-year characteristics that could affect imports. Many other variable are included in typical gravity model specifications. We do not include those variables for two reasons. First, and most obviously, many of the variables in the gravity equation are dyadic (e.g. common language between country i and j), and the analysis conducted here is monadic. Second, as we describe shortly, our preferred specifications all include country-year fixed effects to account for unobserved heterogeneity at the country-year level. The effects of other monadic variables that are often included in a gravity specification (e.g. exchange rates) are captured by the country-year fixed effects in those specifications.

We chose these four country-year variables because we are interested in whether they moderate

[40] Reinhart and Rogoff (2009).

the effects of disputes on imports. We are less concerned with the direct effect of these variables on imports- e.g. whether democracies trade more or less. Those topics are covered extensively elsewhere. As explained in greater detail in the section on responsiveness, we *are* interested in the indirect effects of these variables through their influence on responsiveness, e.g. whether a WTO dispute against a democracy has more or less of an effect on imports. We chose these four country-year characteristics because extant literature gives us theoretical reasons to expect that they moderate the effects of disputes on imports.

All variables are summarized in Table 1. This Table provides summary information for the full sample and then for the sub-sample for which we have full coverage of the country-year control variables. The Table then repeats this content, however, limited only to country-product observations which experience a WTO dispute at some point during the sample.

3 Analysis

Our analysis proceeds in three parts. We first establish baseline results for whether disputes increase trade on average. We then examine whether the effect on trade varies by *dispute* characteristics: the dispute's legal outcome and the relevant issue-area. Lastly, we analyze the effect of dispute settlement looking at *respondent* characteristics, by examining which countries are most responsive to disputes, and why. Overall, examining the WTO's ability to increase trade through dispute settlement, the results are not encouraging.

Before proceeding to our findings, we would note that all of our models include country-product fixed effects to account for unobserved heterogeneity across products and countries. Thus, our tests measure whether there is an increase in post-dispute trade in specific products, relative to mean levels. Since the dispute indicator variable takes on a value of 0 before a dispute and 1 for all years afterwards, this functionally compares years preceding a dispute with years after the dispute, for a

particular country-product series of observations. This approach also means that country-products which never experience a dispute do not influence the estimated coefficient for the dispute variables.

Accounting for unobserved country-product heterogeneity is particularly important because it is well known that disputes target larger product flows, in part because of the large fixed costs associated with pursuing a dispute. Looking at our data, this is indeed the case. Figure 1 shows the values of imports for products that do and do not experience any trade dispute at any point in our sample time frame. The lines show a locally smoothed polynomial for country-products that do experience a dispute (red) and do not ever experience a dispute (blue), along with the accompanying 95% confidence intervals. It is immediately apparent that disputes tend to occur against larger value country-products. Including country-product fixed effects means that we estimate changes in the import values *within* a particular country-product series, not variation *across* country-products. Analyzing variation across products would be misleading in this context because we might mistakenly attribute a positive effect to disputes because of the fact that bigger or faster-growing product lines are more likely to be targeted with disputes.

The inclusion of country-product fixed effects means that the coefficients on the various dispute variables describe a comparison of import values before a dispute to values after a dispute. Since there is no within-country-product variation on the dispute variables for country-products which do not experience a dispute, those observations do not contribute information to the dispute coefficient, though they do contribute information for the coefficient on the country-year control variables.

3.1 Do Disputes Matter? Baseline Results

We start by establishing a pair of baseline results that describe the average effect of WTO disputes on import levels. Specifically, we estimate the following two equations via OLS:

$$IM_{ist} = \beta_0 + \beta_1 * Dispute_{ist} + X_{it}\theta + \alpha_{is} + \epsilon_{ist}$$
$$IM_{ist} = \beta_0 + \beta_1 * Dispute_{ist} + \alpha_{is} + \delta_{it} + \epsilon_{ist} \quad (1)$$

Both specifications analyze whether disputes are associated with increased imports. Both specifications also include country-product fixed effects. They differ only in how they deal with variation at the country-year level, i.e. shocks or covariates that affect a country's trade in all products. The first equation controls for potential sources of country-year level heterogeneity with the matrix of control variables described above (Polity, PC GDP, Respondent, and Crises), X_{it}. The second equation controls for country-year level heterogeneity with country-year fixed effects. We refer to the second specification, with country-year fixed effects, as our preferred specification since it would be difficult to include an exhaustive set of all country-year control variables that affect imports.

The results are presented in Table 2. Looking at the first column, we find encouraging initial results. The coefficient on the binary dispute indicator is positive and significant. The coefficient value of 0.137 indicates that the value of trade flows in years after a dispute is approximately 14% higher in years after a dispute, compared to years before a dispute. The coefficients on the country-year control variables are largely as expected. Richer countries tend to import more, while economic and financial crises decrease imports. Countries targeted with more past disputes tend to import less. Regime type has a negligible impact on imports.

This initial result, however, is not present in the specification which includes country-year fixed effects. While the coefficient on disputes is still positive, it decreases to 0.095 and is not statistically significant, despite the extremely large sample size. Once controlling for unobserved heterogeneity at the country-year level, disputes do not appear to increase trade.

For additional robustness checks, we also ran these same two regressions on separate subsamples. For columns 3 and 4, we excluded the United States and the European Union from the analysis.

Since those two countries are the most involved in disputes, we wanted to be sure that results were not being driven by either of these two litigants. We still get a null result using country-year fixed effects, and additionally, the coefficient on dispute is insignificant even using country-year control variables. Columns 5 and 6 run the regressions on only OECD countries, and columns 7 and 8 use only non-OECD countries. We again see mainly null results. There is a positive and significant coefficient using country-year controls on the non-OECD sample, but this result is still not present when using country-year fixed effects. A large proportion of WTO disputes deal with steel products, so columns 9 and 10 exclude observations pertaining to these products. There is a positive and significant coefficient using country-year controls, but it, too, disappears when using country-year fixed effects.

What about trends in imports before and after disputes? First, it is possible that disputes have a *delayed* effect on imports. After a dispute has occurred, it takes time for the dispute to reach its final resolution. Even once a dispute it resolved, it could take time for market actors to readjust their production and consumption behavior sufficiently to see an increase in imports. To examine this possibility, we performed two additional sets of analysis. First, we re-estimated the first two columns in Table 3 using lagged dispute variables. These regressions are designed to detect increases in imports that could be delayed from one to six years. While most of the coefficients on various lagged dispute variables and specifications are positive, they are all statistically insignificant. If it takes time for disputes to have a positive effect on trade, this effect is not statistically apparent even six years after the dispute's initiation.

For a second piece of evidence, we use non-parametric analysis to describe the effects of disputes on imports. Figure 2 shows the distribution of import values for disputed products, according to the years before and after a dispute. Year zero corresponds to the year that the dispute reached its outcome. The lines are the smoothed values from a local regression of imports on the year value.

The top pane examines the ten years before and after a dispute and the bottom pane "zooms in" on the five years preceding and following a dispute. If disputes had a positive effect on imports, we would expect to see an increase in import values near or immediately after year zero. Instead, import values of disputed products tend to increase steadily in the years preceding and following a dispute, and the dispute itself appears to have no effect on the upward trend in import values.

The smoothed lines also do not display strong evidence of a delayed effect of disputes. There are not strong upward "kinks" after year zero, which would indicate a delayed positive effect of disputes. There is a slight increase in the slope of the line in the 4th and 5th years after a dispute, but as demonstrated with the lagged dispute variable regressions, it falls short of statistical significance.

It is also possible that there are trends in imports *before* a dispute that bias us against finding positive dispute effects in our regression specifications. For example, consider a dispute over an antidumping duty. In the canonical antidumping situation, we would expect to see an initial surge of imports (resulting from the dumping), then a decrease (resulting from the antidumping duty), and then a subsequent increase in imports after the WTO dispute potentially resulted in the removal of antidumping duties. Since our regression specification compares variation in pre- and post-dispute import levels within particular country-products, the initial surge in imports might artificially bias the coefficient on the dispute variable downward. While this is plausible, we do not see evidence of increases in imports before disputes. Looking again at Figure 2, there are not distinct increases in imports, above mean levels, before disputes. In the top pane, imports decrease and then increase a long time 5-10 years before a dispute. In the bottom pane, there is very little variation at all.

3.2 Does Dispute Outcome Matter?

Next, we re-estimate the first two models, but instead of the simple binary dispute indicator, we include a series of binary variables that indicate the status of a particular dispute (pre-dispute, dispute ongoing, MAS, withdrawn, and panel), as described above. We had prior theoretical reasons to expect that all disputes, regardless of outcome, should affect trade. Yet, the null results from the preceding section suggest that the relationship between disputes and trade may depend on whether, for example, a ruling is issued as opposed to a mutually agreed solution. For each of the last four statuses (dispute ongoing, MAS, withdrawn, and panel), let s^j_{ist} indicate whether country-product is was in that stage of the dispute during year t. Including these status indicators allows us to use the pre-dispute status as a base category. The two econometric models estimated here are:

$$
\begin{aligned}
IM_{ist} &= \beta_0 + \sum_{j=1}^{4} \beta_j * s^j_{ist} + X_{it}\theta + \alpha_{is} + \epsilon_{ist} \\
IM_{ist} &= \beta_0 + \sum_{j=1}^{4} \beta_j * s^j_{ist} + \alpha_{is} + \delta_{it} + \epsilon_{ist}
\end{aligned}
\tag{2}
$$

Table 4 shows the results from a regression of import values on the status indicators described above. As in Table 1, the first column includes country-year control variables and the second column includes country-year fixed effects. Only withdrawn—or, "dropped"—disputes are consistently associated with increases in imports after the dispute. In both specifications, years after a dropped dispute experience approximately 25% higher import values, and this result is statistically significant. The results here would be consistent with the argument that complainants drop their claims because respondents have dismantled their offending entry barrier and allowed greater

access to the market.

Importantly, disputes that reach the panel stage and disputes that end in a mutually agreed solutions are not associated with significant increases in import values. On the one hand, this is surprising. Mutually agreed solutions and panel rulings are the two events that are most closely associated with a thriving dispute settlement mechanism. Yet according to the findings, they are also associated with weak increases in trade.

In the case of panel rulings, the null finding is less surprising once we consider that disputes that reach the panel stage are likely to be those involving the hardest legal issues. After all, the reason a dispute reaches the panel stage in the first place is because of the parties' inability to resolve the dispute bilaterally. Thus, in spite of the expectation that rulings have the sharpest teeth, disputes that reach the panel stage may be those that are least ripe for increases in trade volume.

The null finding on MAS is more surprising, given that mutually agreed solutions are, by definition, a compromise between the complainant and respondent. However, there may also be good reason for the null result for mutually agreed solutions. If the two parties negotiate a narrowly tailored settlement, especially one that only benefits the complainants, possibly even at the expense of parties not in the room,[41] then we might not expect to see large increases in trade on average. While the DSU's rules specifically proscribe settlements that disproportionately favor the parties in the room, this rule is difficult to enforce. Indeed, the way by which the DSU encourages settlement is to ensure the privacy of consultations, and of the resulting early settlements. Such privacy allows the parties greater insulation from pressures from powerful domestic groups, but it is also likely to increase the odds of discriminatory settlements. It may be that the null finding for MAS is

[41] While most such settlements are by construction most likely to be unobservable, we have enough anecdotal evidence to warrant concern. Consider a dispute brought by Korea against Japanese restrictions on seaweed imports, *Japan—Quotas on Laver*, where Japan effectively allocated its entire quota of dried and seasoned laver imports exclusively to South Korea. As the settlement read: "An annual import quota will be allocated exclusively for Korean Laver Products." WT/DS323/5. Observers denounced this agreement as discriminating against other members to the benefit of the parties in the room (Nakagawa, 2007).

evidence for this. In either case, the empirical result is unambiguous: early settlement does not lead to significant increases in trade for the membership at large.[42]

There is some evidence that import values increase as disputes are ongoing. In the country-year controls specification, years during which the dispute is ongoing are associated with approximately 12% higher import values, though this estimate drops to approximately 7% in the country-year fixed effects specification, and ceases to be statistically significant. As a result, it is difficult to draw the conclusion that ongoing disputes are robustly associated with an increase in trade.

These same results obtain when using the same subsamples examined above, excluding US/EU, OECD only, non-OECD only, and excluding steel. Withdrawn cases are associated with significant increases in trade in all but one subsample and specification (non-OECD only, with country-year fixed effects). Disputes that end in a mutually agreed upon solution are not associated with increased trade in any of the models. Panel rulings have a positive, significant coefficient only when looking at non-OECD countries. The slightly positive results for ongoing disputes become weaker when looking across subsamples, with the coefficient even turning negative in two specifications.

Again, nonparametric analysis comports with the regression results. Figure 3 shows the smoothed values of trade, before and after a dispute ends, broken down by whether the dispute ended in withdrawal, a MAS, or a panel.[43] The blue lines, corresponding to withdrawn disputes, shows how import values increase after these disputes, though these values were also increasing prior to the conclusion of the dispute. The green lines, corresponding to disputes that reached the panel stage, show modest and more gradual increases in import values. The red lines, corresponding to mutually agreed upon solutions, show virtually no increases in trade in the years preceding or following the conclusion of these disputes.

[42] Note that the coefficients for the control variables are identical to the previous table. This is because the status variables simply break down the dispute variable into particular components. These regressions do not introduce any new variation that is correlated with the control variables.

[43] This figure uses the same techniques from Figure 2.

3.3 Does Issue-Area Matter?

We now explore whether the efficacy of disputes is conditioned by the content of the case. To examine whether issue-area matters, we re-estimate the models in equation 1, but now include indicator variables for each of the issue-areas described above. Table 5 shows results from a regression of imports on the issue-area indicators. Once again, we include country-year controls and country-year fixed effects, in columns 1 and 2, respectively.

WTO disputes are more successful at increasing imports for some issue-areas than for others. In both specifications, safeguards disputes are associated with a positive and significant increase in post-dispute imports. On the other hand, antidumping disputes have, at best, a negligible effect on imports. In the second model, antidumping disputes have actually been associated with an approximate 18% *decrease* in imports.[44] This result in particular is surprising and discouraging, since antidumping disputes constitute a significant portion of the DSU's caseload. However, as pointed out above, antidumping is often used because of the domestic political pressure to insulate the marketplace from competitive pressures. Thus, the high frequency of antidumping use means it is more likely to be targeted under the DSU. Yet, the domestic politics of trade protection means respondents are unwilling to give ground over the course of a WTO dispute. The troubling implication is that the DSU is not promoting trade in cases involving some of the most salient and heavily relied on non-tariff barriers.

The empirical appendix further demonstrates the robustness of these results. We estimated our models using the year that disputes concluded, rather than began, to ensure that the years a dispute was ongoing did not bias our results for the effect of a dispute on imports downward. We also restricted our sample in several ways. One restriction was to look separately at disputes ending before the years 2000, 2002, and 2005 in order to make sure our results were not driven by

[44] Adding the dispute and AD coefficients, $-0.251 + 0.078$.

country-products with too few post-dispute observations. Another was to restrict the sample by the number of products covered by the dispute so that disputes targeting larger or smaller numbers of goods were not driving results. We then reran our models looking only at disputes that involved import measures, excluding those (approximately 20) disputes that involved policies designed to promote exports. Since our outcome of interest is imports into the respondent market, we wouldn't necessarily expect a dispute against these policies to affect imports. Finally, we explored different windows of time *prior* to disputes. Since our results rely chiefly on a comparison of pre- and post-dispute import levels, this helps confirm that the estimates were not an artifact of the reference period. All of these robustness checks produced estimates consistent with the models reported here. Under no conditions are disputes robustly associated with a significant increase in trade.

3.4 Does the Respondent Matter?

In this section, we analyze which countries are the most responsive to WTO disputes? For a variety of reasons, certain members may be willing to give more ground when taken to dispute settlement than others. If members vary systematically in their willingness to dismantle their entry barriers, it will be difficult to identify whether disputes have a positive effect on trade. We show that *who* one sues matters at least as much as what one sues over and the outcome of the dispute.

To analyze which countries are most responsive to WTO disputes, we estimate that following model:

$$\begin{aligned} IM_{ist} &= \beta_0 + \beta_i * Dispute_{ist} + \alpha_{is} + \delta_{it} + \epsilon_{ist} \\ \beta_i &= u_{1i} \\ \alpha_{is} &= u_{2is} \\ \delta_{it} &= u_{3it} \end{aligned} \tag{3}$$

Importantly, this model allows us to estimate a country-specific component of the coefficient on our dispute indicator variable, u_{1i}. This model also affords us a great deal of flexibility, since we can estimate a country-product-specific intercept, u_{2is}, and a country-year-specific intercept, u_{3it}, similar to the fixed effects models estimated in the previous sections.

Figure 4 plots the country-specific dispute coefficient, u_{1i} estimated from the model.[45] The results are striking for two reasons. First, there is a lot of variation in the degree to which countries are responsive to WTO disputes. On the one hand, countries like India are very responsive, with an estimated dispute coefficient of over 0.5. On the other hand, disputes against some countries have actually been associated with *decreased* trade flows. For example, the estimated dispute coefficients for Korea and Mexico are approximately -0.55 and -0.40, respectively. The second striking feature of the result is that a large number of country coefficients are clustered around zero, meaning these countries have largely been unresponsive to trade disputes. 11 of the 31 countries have dispute coefficients that are between -0.15 and 0.15.

3.4.1 What Explains Responsiveness?

Given the significant variation in countries' responsiveness to dispute settlement, we now ask: are there systematic factors driving the difference in responsiveness across members? We focus on variation at the country-year level, i.e. what country-year characteristics increase or decrease that country's responsiveness to WTO disputes? For consistency, we use the set of country-year control variables described in previous sections. These are income, regime type, previous dispute experience, and whether a domestic market is experiencing an economic crisis. Each of these can be reasonably expected to condition a state's likelihood of granting market access as a result of a dispute.

To estimate the effects of these country-year characteristics on how responsive a country is to

[45] The multilevel models were estimated with the lme4 package in R (Bates et al., 2014).

WTO disputes, we use the following multilevel model:

$$\begin{aligned} IM_{ist} &= \beta_0 + \beta_{it} * Dispute_{ist} + \alpha_{is} + \delta_{it} + \epsilon_{ist} \\ \beta_{it} &= b_0 + b_1 * Polity_{it} + b_2 * lnGDPPC_{it} + b_3 * Resp._{it} + b_4 * Crisis_{it} + u_{1it} \\ \alpha_{is} &= u_{2is} \\ \delta_{it} &= d_1 * Polity_{it} + d_2 * lnGDPPC_{it} + d_3 * Resp._{it} + d_4 * Crisis_{it} + u_{3it} \end{aligned} \tag{4}$$

The key feature of this model is that it allows for the coefficient on the dispute indicator, β, to be influenced by country-year characteristics. Looking at the second line of the model, b_1-b_4 describe whether each of the four moderating variables increases or decreases a country's responsiveness to trade disputes. The third and fourth line describe country-product and country-year random effects.

The second line of the model is akin to the more familiar interactive term approach, in which the dispute variable would be interacted with each of the four moderating variables. The key difference is that this approach does not force the effect of a dispute to be solely determined by the four moderating variables. Rather, the effect of a dispute also has a country-year specific component, outside of the effect of the four moderating variables, u_{1it}. To see the difference between this model and its interactive term cousin, substitute the second line into the first. If u_{1it} equals 0, substituting the second lines into the first line yields an equation identical to the multiplicative interaction term model with dispute interacted with each of the country-year variables. When $u_{1it} \neq 0$, as is very likely be the case since our four moderating variables are almost assuredly not the *only* four variables that influence dispute responsiveness, using the multilevel model is important. The multilevel model does not require the assumption that β_{it} precisely equals $b_0 + b_1 * Polity_{it} + b_2 * lnGDPPC_{it} + b_3 * Resp._{it} + b_4 * Crisis_{it}$ precisely.

Another ancillary benefit of the multilevel model is that results—specifically, the b's from the

second line—have a straightforward interpretation. If b_k is positive, for any of the k moderators, then an increase in that variable makes a country more responsive to a WTO dispute. If b_k is negative, an increase in that moderator makes a country less responsive to WTO disputes.

The results are presented in Table 6. The top part of the table shows how each of the four country-year variables moderates the effect of a dispute on imports. Some of these estimates are signed as expected. Countries suffering from more economic and financial crises are less responsive to WTO disputes. WTO disputes against countries with more crises have a less positive effect on imports, as indicated by the negative estimate for *Crisis*. This result is consistent with the idea that countries will protect their markets more vehemently during crises, regardless of whether they are sued in the global trade regime.

Similarly, richer countries and countries that have been more frequently targeted with disputes in the past are less responsive to WTO disputes, as indicated by the negative estimates for *lnpcgdp* and *Respondent*. Disputes targeting poorer countries and countries that have not had to defend themselves against as many WTO disputes in the past are associated with greater increases in imports. Somewhat unexpectedly, more democratic countries are significantly less responsive to trade disputes. However, while the traditional story about democratic audiences is that they punish violations of international law, we have stressed that WTO-illegal entry barriers are a direct result of (at least some portion of) audiences demanding such violations. In the case of dispute settlement, therefore, democratic audiences may punish their governments for backing down and exposing the domestic marketplace to costly foreign competition.

4 Conclusions

This paper presents a striking finding: the effect of WTO disputes on the imports at issue is, at best, inconsistent. Disputes are not associated with a statistically significant increase in aggregate

imports, compared with prior to the dispute. Looking at particular dispute outcomes and issue areas, only certain types of disputes have been associated with increased trade, while others have resulted in decreases. Moreover, certain respondents are more "responsive" to disputes than others. Disputes against some countries have resulted in increased imports, while disputes against others have failed to do so. We do not find strong evidence of a systematic explanation for this variation.

As students of the WTO seeking to confirm our priors about the effects of dispute settlement on trade, we are admittedly surprised to discover that WTO disputes are not robustly associated with increased trade, something which the literature—our own past work included—has taken largely for granted. Before considering the discrepancy between our findings and the prevalent view of dispute settlement as the crown jewel of the trade regime, it is worth addressing some possible empirical concerns.

First, perhaps the lack of a positive finding can be chalked up to the monadic nature of our analysis, and a dyadic analysis may be better at unearthing a positive effect at the product level. Perhaps WTO disputes benefit complainants and/or third parties, but not the membership at large. Answering both questions—the effect of disputes on imports into the respondent from the complainant *and* the effect on imports from all other members—is key to understanding the overall effects of the DSU. Our concern has been to show whether dispute settlement leads to an observable increase in trade, i.e. the "size of the pie." Dyadic analyses could find that disputes increase trade for dispute participants compared to nonparticipants, and at least one study does this. The risk, however, is misconstruing trade diversion as increased market access. To put the matter in stark terms, if dispute settlement is increasing market access for certain countries, but not increasing the total amount of imports, then a positive dyadic finding is not necessarily good news for the DSU. Rather than increasing the size of the pie, disputes may be just re-allocating of each members' slice of the pie. Dyadic analysis at the product level is an important next step.

Second, this paper necessarily focused on "the dogs that bark," i.e. disputes that actually occurred, and not the dogs that didn't bark: potential disputes that could have occurred but did not. One implication is that WTO disputes may be deterring countries from implementing certain trade barriers. If this is the case, then countries may not be filing disputes to liberalize trade in the product or even the country they are challenging, but they may instead be sending a warning to the broader membership about the legal measure at issue, and their willingness to take it on in dispute settlement. This is entirely plausible. If so, future research would need to look for the impact of dispute settlement across all traded products, even those not involved in the dispute, since deterrence or precedential value would be expected to have a diffuse effect. The mere possibility of a WTO dispute might deter a country from implementing a harmful protectionist barrier in the first place. Alternatively, a dispute between countries i and j might deter country k from erecting trade barriers. A dispute against country i over product s might deter country i from erecting trade barriers for another product, t. The possible channels are complex and not mutually exclusive. Examining whether, and under what conditions, WTO disputes deter trade barriers is of paramount importance, but is beyond the scope of this paper.

What, then, to make of our main finding, that the average WTO dispute yields no observable increases in trade? While we have focused on the most natural place to look for an effect—trade in the product at issue in the dispute, following its conclusion—it may be that most of the impact of dispute settlement is felt elsewhere. It may be, for instance, that the benefit of dispute settlement rests entirely on its rule-clarifying function. In such a telling, a dispute's resolution leads to a convergence of expectations over country behavior. If most noncompliance arises from a misunderstanding of the meaning of rules,[46] the main benefit of a dispute may be to clarify the meaning of the underlying law. This benefit would be felt throughout the institution, as countries adjust their behavior to take into account that, e.g. the use of safeguards is now known to require

[46] Chayes and Chayes (1993).

the demonstration of "unforeseen developments".[47] If this is the case, however, our null finding over the effect on trade of mutually agreed solutions is of special concern. Indeed, settlements have no clarifying effect, since their content is usually kept largely private. When taken together with the finding that settlements have no average effect on trade, there remains little to recommend settlements. This is especially striking given the literature's contention that MAS are where most of the action takes place in dispute settlement.

Related to the idea of rule-clarification, it could be that dispute settlement is mostly an exercise in changing the meaning of the rules to one's advantage. Perhaps countries file to set favorable precedents about matters of concern to them. Although the WTO features no formally binding precedent, or *stare decisis*, scholars largely agree that something akin to *de facto stare decisis* actually takes place,[48] where rulings do constrain the rulings of judges in subsequent cases. In this way, anecdotal evidence suggests that in the early years of the WTO, the United States filed a number of low-stake cases over intellectual property rights as a means of solidifying the interpretation of the recently passed TRIPS agreement. In other words, rule clarification may be strategic, if its outcome favors one country over another.

It may be that since disputes take time—just under two years on average—this offers enough respite to domestic producers to improve their productivity. Once the barriers are taken down, foreign exporters would not capture as much of the market as they would have before the dispute. More pessimistically, perhaps respondents simply replace one barrier with another, giving credence to Bhagwati's law of constant protection. There exists anecdotal evidence to this effect; the findings may suggest that such a bait-and-switch is actually the norm.

While these alternative mechanisms could indeed be at work, most scholars have long assumed

[47] Starting in 1997, the AB began enforcing the long-ignored "unforeseen developments" provision clause which dated back to the GATT : "[w]e have recently confirmed this principle in our Report in [*Korea–Dairy*]" (*Argentina Safeguard Measures on Imports of Footwear*, WT/DS121/AB/R, fn. 72).

[48] Bhala (1998).

that countries file disputes to obtain market access for domestic industries, and that under MFN rules, concessions by the respondent are extended to the remainder of the membership. In this way, other countries, and especially developing countries, are thought to free-ride on the litigation efforts of (usually wealthier) states. We do not find strong support for this view. There is more work to be done to uncover the conditions under which disputes increase trade.

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Table 1: Summary Statistics

	All Country-Products						Country-Products with Dispute					
	Full Sample			with Controls			Full Sample			with Controls		
Variable	Mean	S.D.	N	Mean	S.D.	N	Mean	S.D.	N	Mean	S.D.	N
Log Imports	13.45	3.13	2,228,653	13.52	3.05	1,606,257	14.66	3.16	109,918	14.56	3.13	83,035
Dispute	0.03	0.18	2,228,653	0.04	0.20	1,606,257	0.69	0.46	109,918	0.81	0.40	83,035
Polity				6.86	4.28	1,606,257				8.26	2.99	83,035
ln(PC GDP)				8.32	1.16	1,606,257				9.24	1.20	83,035
Respondent				1.58	1.02	1,606,257				2.52	1.09	83,035
Crisis				0.86	1.14	1,606,257				0.94	1.14	83,035

Summary statistics for dependent and explanatory variables. The first six columns summarize the sample for all country-product observations. The second six columns summarize the sample for all country-products that experience at least one dispute.

Table 2: Effect of WTO Disputes on Imports

	Controls	FE	No US-EU	No US-EU	OECD	OECD	nonOECD	nonOECD	NoSteel	NoSteel
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dispute	.137	.095	.132	.132	-.021	.065	.231	.078	.135	.097
	(.076)*	(.095)	(.089)	(.116)	(.057)	(.116)	(.119)*	(.173)	(.080)*	(.101)
Polity	-.012		-.012		-.015		-.012		-.013	
	(.009)		(.009)		(.084)		(.008)		(.009)	
ln(PC GDP)	2.847		2.859		2.567		2.957		2.793	
	(.421)***		(.411)***		(.494)***		(.465)***		(.406)***	
Respondent	-.185		-.200		-.057		-.236		-.182	
	(.037)***		(.037)***		(.069)		(.031)***		(.038)***	
Crisis	-.039		-.042		-.031		-.039		-.039	
	(.019)**		(.019)**		(.017)*		(.023)*		(.019)**	
N	1,606,257	2,228,653	1,536,050	2,087,814	467,135	630,809	1,139,122	1,597,844	1,508,124	2,093,198
No. Products	129,545	149,216	124,623	139,397	34,183	39,077	99,666	143,866	121,869	140,395
N (w/ disp.)	83,035	109,918	63,773	81,125	54,163	58,728	28,872	51,190	78,054	100,573
No. Prod. (w/ disp.)	5,921	6,362	4,602	4,623	3,845	4,265	2,586	5,922	5,584	5,745

Results from OLS regression of logged import values on dispute status and various fixed effects combinations. Each regression uses robust standard errors clustered at the country level.

Table 3: Effect of Lagged WTO Disputes on Imports

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$Disp._{t-1}$	.102	.087										
	(.080)	(.089)										
$Disp._{t-2}$			.070	.088								
			(.079)	(.077)								
$Disp._{t-3}$					.031	.075						
					(.080)	(.070)						
$Disp._{t-4}$							.029	.050				
							(.086)	(.064)				
$Disp._{t-5}$									.003	.018		
									(.100)	(.060)		
$Disp._{t-6}$											-.047	-.015
											(.124)	(.057)
Polity	-.012		-.012		-.012		-.011		-.010		-.008	
	(.009)		(.009)		(.009)		(.008)		(.008)		(.008)	
ln(PC GDP)	2.845		2.845		2.845		2.929		3.138		3.301	
	(.421)***		(.421)***		(.422)***		(.414)***		(.336)***		(.282)***	
Respondent	-.184		-.183		-.182		-.213		-.264		-.284	
	(.037)***		(.037)***		(.037)***		(.039)***		(.056)***		(.071)***	
Crisis	-.039		-.039		-.039		-.033		-.021		-.023	
	(.019)**		(.019)**		(.019)**		(.020)*		(.020)		(.017)	
N	1,606,257	2,155,821	1,606,257	2,074,688	1,606,257	1,978,385	1,555,644	1,877,994	1,474,646	1,770,796	1,368,881	1,660,351
No. Products	129,545	149,186	129,545	149,131	129,545	149,049	129,301	148,628	129,142	148,367	128,920	148,146
N (w/ disp.)	83,035	105,567	83,035	100,441	83,035	95,137	79,165	89,797	73,651	84,206	68,036	78,578
No. Prod. (w/ disp.)	5,921	6,362	5,921	6,362	5,921	6,362	5,920	6,359	5,919	6,358	5,914	6,353

Results from OLS regression of logged import values on dispute status and various fixed effects combinations. Each regression uses robust standard errors clustered at the country level.

Table 4: Effect of WTO Disputes on Imports, by Dispute Status

	Ctrls	FE	No US-EU	No US-EU	OECD	OECD	nonOECD	nonOECD	NoSteel	NoSteel
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dropped	.251	.271	.230	.244	.220	.268	.287	.197	.226	.244
	(.098)**	(.098)***	(.110)**	(.123)**	(.075)***	(.089)***	(.139)**	(.184)	(.105)**	(.120)**
MAS	.054	-.022	.029	.045	-.084	-.073	.116	-.035	.062	-.010
	(.128)	(.132)	(.166)	(.205)	(.077)	(.064)	(.258)	(.320)	(.126)	(.133)
Panel	.141	.074	.135	.112	-.053	.075	.385	.143	.155	.107
	(.107)	(.133)	(.128)	(.163)	(.097)	(.182)	(.118)***	(.084)*	(.123)	(.146)
Ongoing	.123	.074	.125	.114	-.027	.125	.186	-.016	.137	.089
	(.067)*	(.098)	(.071)*	(.118)	(.048)	(.133)	(.100)*	(.126)	(.070)*	(.102)
Polity	-.012		-.012		-.016		-.012		-.013	
	(.009)		(.009)		(.084)		(.008)		(.009)	
ln(PC GDP)	2.844		2.857		2.566		2.952		2.790	
	(.420)***		(.411)***		(.495)***		(.464)***		(.405)***	
Respondent	-.184		-.199		-.055		-.234		-.180	
	(.037)***		(.037)***		(.069)		(.031)***		(.038)***	
Crisis	-.039		-.042		-.031		-.039		-.039	
	(.019)**		(.019)**		(.017)*		(.023)*		(.019)**	
N	1,606,257	2,228,653	1,536,050	2,087,814	467,135	630,809	1,139,122	1,597,844	1,508,124	2,093,198
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N (w/ disp.)	83,035	109,981	63,773	81,125	54,163	58,728	28,872	51,190	78,054	100,573
No. Prod. (w/ disp.)	5,921	6,362	4,602	4,623	3,845	4,265	2,586	5,922	5,584	5,745

Results from OLS regression of logged import values on dispute status. The omitted category is status 0, meaning years before a dispute has been initiated. Each regression uses robust standard errors clustered at the country level.

Table 5: Effect of WTO Disputes on Imports, by Dispute Issue Areas

	(1)	(2)
Dispute	.130	.078
	(.088)	(.130)
AD	-.130	-.251
	(.117)	(.112)**
Safeguards	.140	.286
	(.056)**	(.060)***
Agriculture	.092	.018
	(.081)	(.111)
SPS	-.275	.020
	(.174)	(.185)
Polity	-.012	
	(.009)	
lnpcgdp	2.848	
	(.421)***	
Respondent	-.185	
	(.038)***	
Crisis	-.039	
	(.019)**	
Country-prod. FE	Y	Y
Country-year FE	N	Y
N	1,606,257	2,228,653

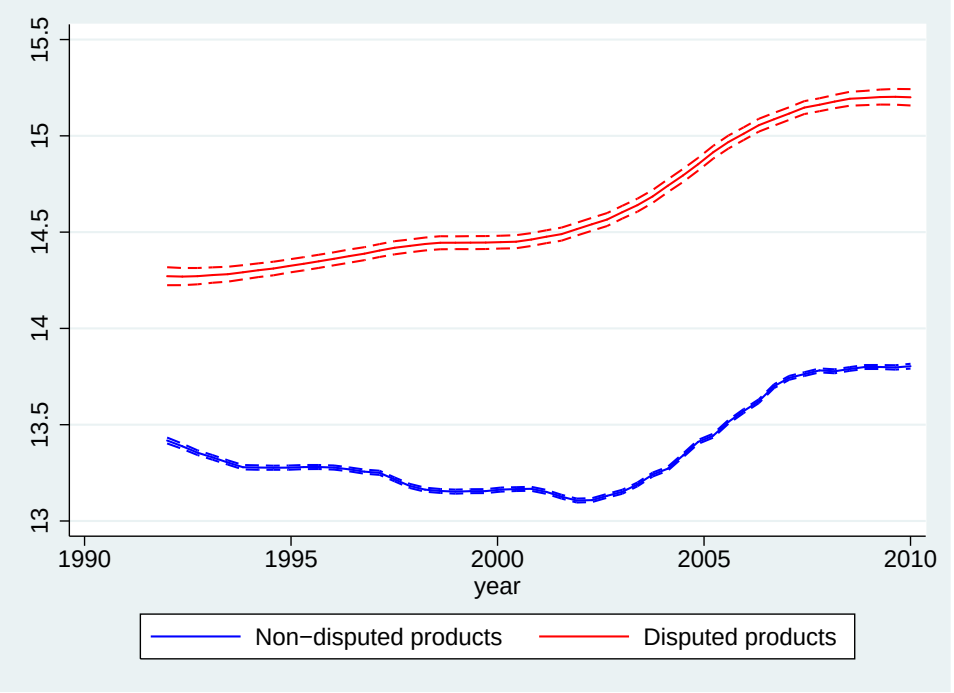
Results from OLS regression of logged import values on dispute status. Each regression uses robust standard errors clustered at the country level.

Table 6: Multilevel Model of Effect of WTO Disputes on Imports

		Fixed Effects (b's)			
Variable	Estimate	S.E.	t value		
Polity	-0.036553	0.009654	-3.79		
lnpcgdp	-0.149259	0.033572	-4.45		
Respondent	-0.097087	0.049612	-1.96		
Crisis	-0.058052	0.027690	-2.10		
Intercept	13.323052	0.085642	155.57		
Dispute	1.702281	0.262342	6.49		
		Correlation of Fixed Effects			
	Int.	Dispute	Polity	lnpcgdp	Respondent
Dispute	0.035				
Polity	-0.001	0.142			
lnpcgdp	-0.002	-0.883	-0.296		
Resp.	0.001	-0.021	-0.174	-0.357	
Crisis	-0.002	-0.116	-0.044	0.036	0.024
N	393971				
Country-Products	117752				
Country-Years	374				

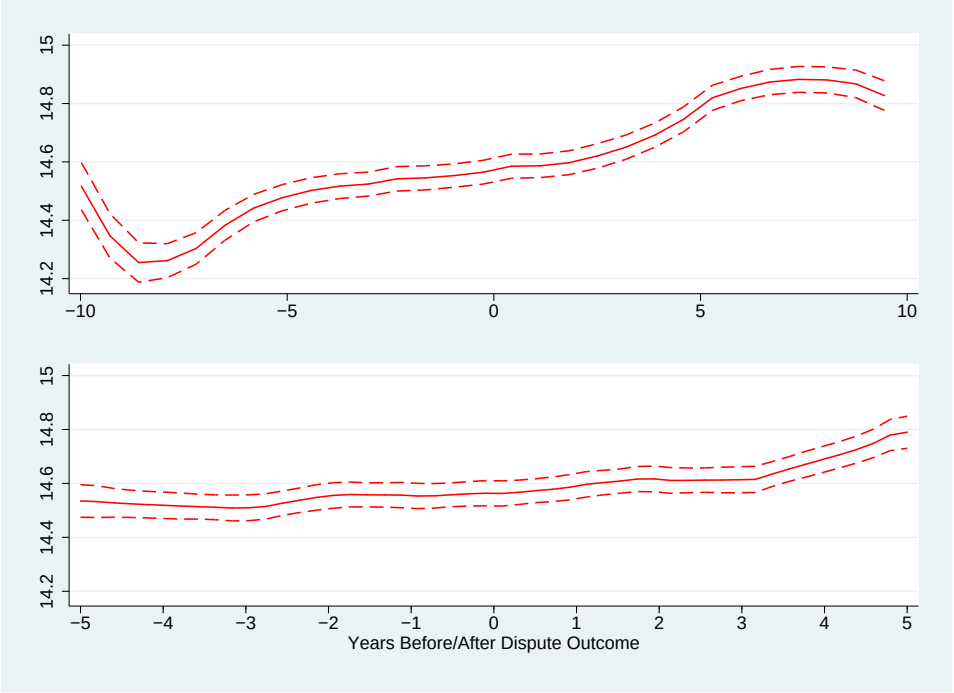
Results from restricted maximum likelihood estimation of Model 4.

Figure 1: Import Values Over Time, Disputed and Undisputed Products



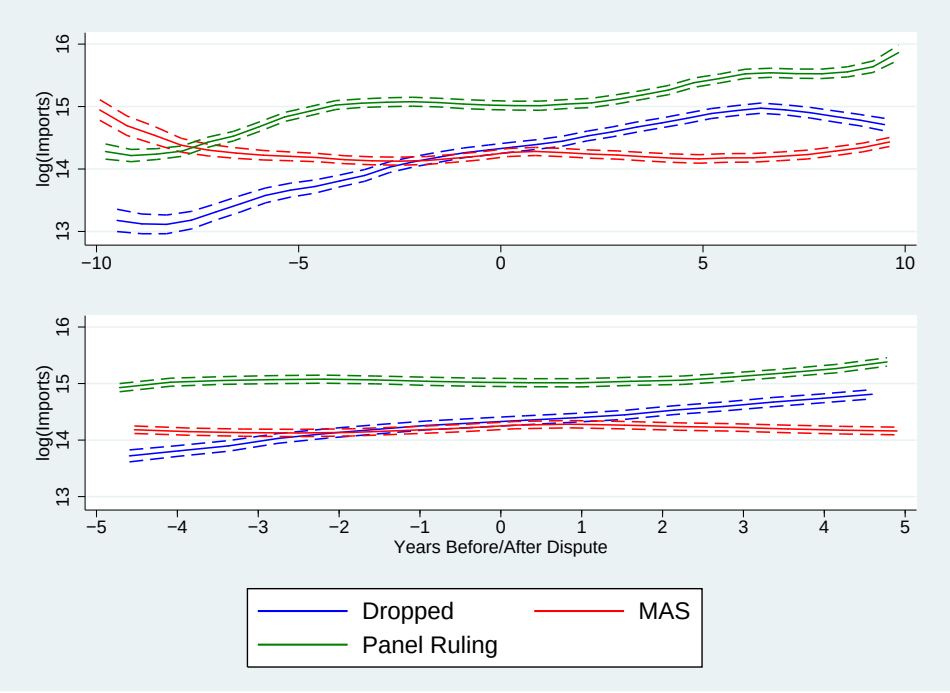
Value of log(imports) for country-products that were disputed verses undisputed, with 95% confidence intervals.

Figure 2: Import Values by Year Before and After Dispute Conclusion



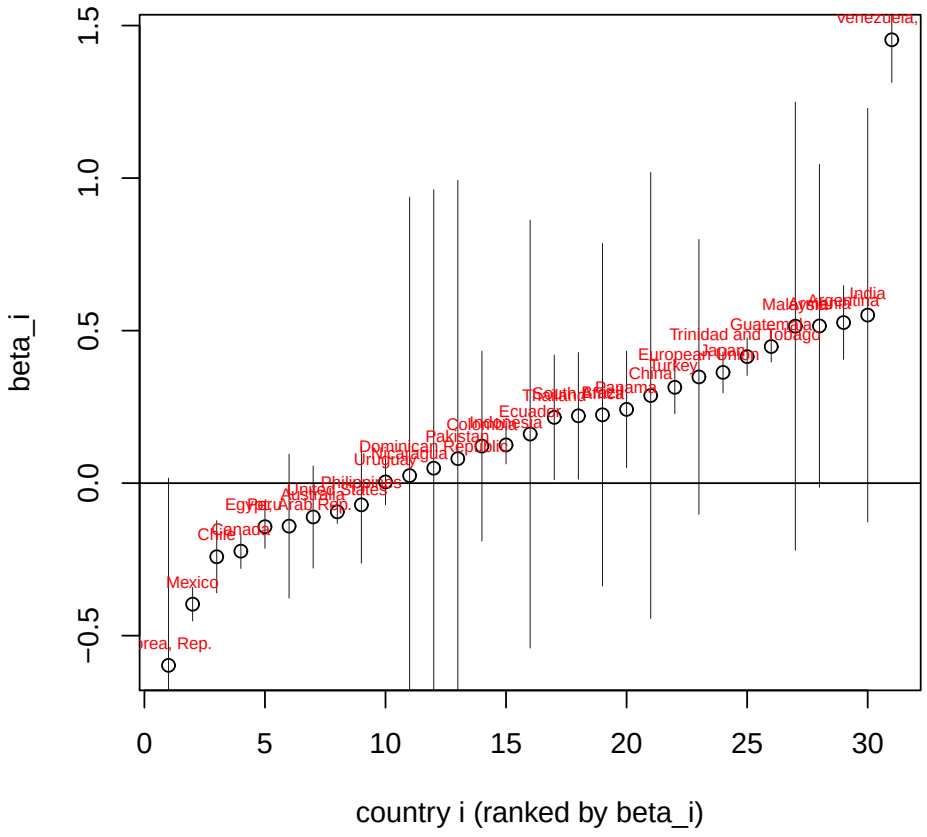
Smoothed value of log(imports) for country-products. Top pane is for 10 years prior to and after a dispute's outcome. The bottom pane is for plus or minus five years. Dashed lines are 95% confidence intervals.

Figure 3: Import Values by Year Before and After Dispute Conclusion, for Dropped, MAS, and Panel Ruling Disputes



Smoothed value of log(imports) for country-products. Top pane is for 10 years prior to and after a dispute's outcome. The bottom pane is for plus or minus five years. Dashed lines are 95% confidence intervals.

Figure 4: Country Specific Effects of WTO Disputes



The vertical axis shows the estimated country-specific dispute coefficient and the 95% confidence intervals around that estimate. Horizontal axis shows the country's rank, in terms of their estimated responsiveness coefficient.